

Research Publications

Peer-Reviewed Journal Articles

1. **Q. H. Nguyen**, O. Sadki, H. Rafaralahy, M. Haddad, A. Zemouche, *Cyberattack Detection by Using a Discrete-Time Model-Based Unknown Input Observer*, IEEE Control Systems Letters, vol. 8, pp. 856–861, 2024.

International Conference Papers

1. **Q. H. Nguyen**, S. Meng, M. Haddad, H. Rafaralahy, A. Zemouche, *Neural Observer-Based Learning for Robust State Estimation*, International Conference on Electrical and Computer Engineering (ICEECE), Madagascar, 2025.
2. S. Bougherara, **Q. H. Nguyen**, M. Haddad, H. Rafaralahy, A. Zemouche, *Application of Discrete-Time Unknown Input Observers for Cyberattack Detection in Connected CACC Vehicles*, International Conference on Electrical and Computer Engineering (ICEECE), Madagascar, 2025.
3. S. Meng, **Q. H. Nguyen**, A. Zemouche, F. Meng, F. Zhang, *Distributed High-Gain Observer for Nonlinear Connected Autonomous Vehicles*, American Control Conference (ACC), 2025.
4. O. Sadki, **Q. H. Nguyen**, A. Zemouche, H. Rafaralahy, M. Haddad, *Cyberattack Estimation in Intelligent Transportation Systems Using an Adaptive High-Order Sliding Mode Differentiator*, 33rd Mediterranean Conference on Control and Automation (MED), 2025.
5. **Q. H. Nguyen**, O. Sadki, H. Rafaralahy, M. Haddad, A. Zemouche, *Cyberattack Detection by Using a Discrete-Time Model-Based Unknown Input Observer*, IEEE Conference on Decision and Control (CDC), 2024.

Submitted and In-Preparation Manuscripts

1. **Q. H. Nguyen**, S. Meng, M. Haddad, H. Rafaralahy, A. Zemouche, *Resilient Trust-Aware Distributed Observer Design for Connected Vehicle Platoons*, 23rd IFAC World Congress, 2026 (submitted).
2. S. Meng, **Q. H. Nguyen**, M. Alma, A. Zemouche, M. Haddad, *Distributed Unknown Input Observer for Vehicle Platoons Under Sensor Faults*, 23rd IFAC World Congress, 2026 (submitted).
3. **Q. H. Nguyen**, S. Meng, M. Haddad, H. Rafaralahy, A. Zemouche, *Control-Oriented Trust-Aware Observer Design for Connected Vehicle Platoons*, IEEE Transactions on Intelligent Transportation Systems (in preparation).

Poster Presentation

1. **Q. H. Nguyen**, O. Sadki, H. Rafaralahy, M. Haddad, A. Zemouche, *Cyberattack Detection by Using a Discrete-Time Model-Based Unknown Input Observer*, American Control Conference (ACC), 2024.